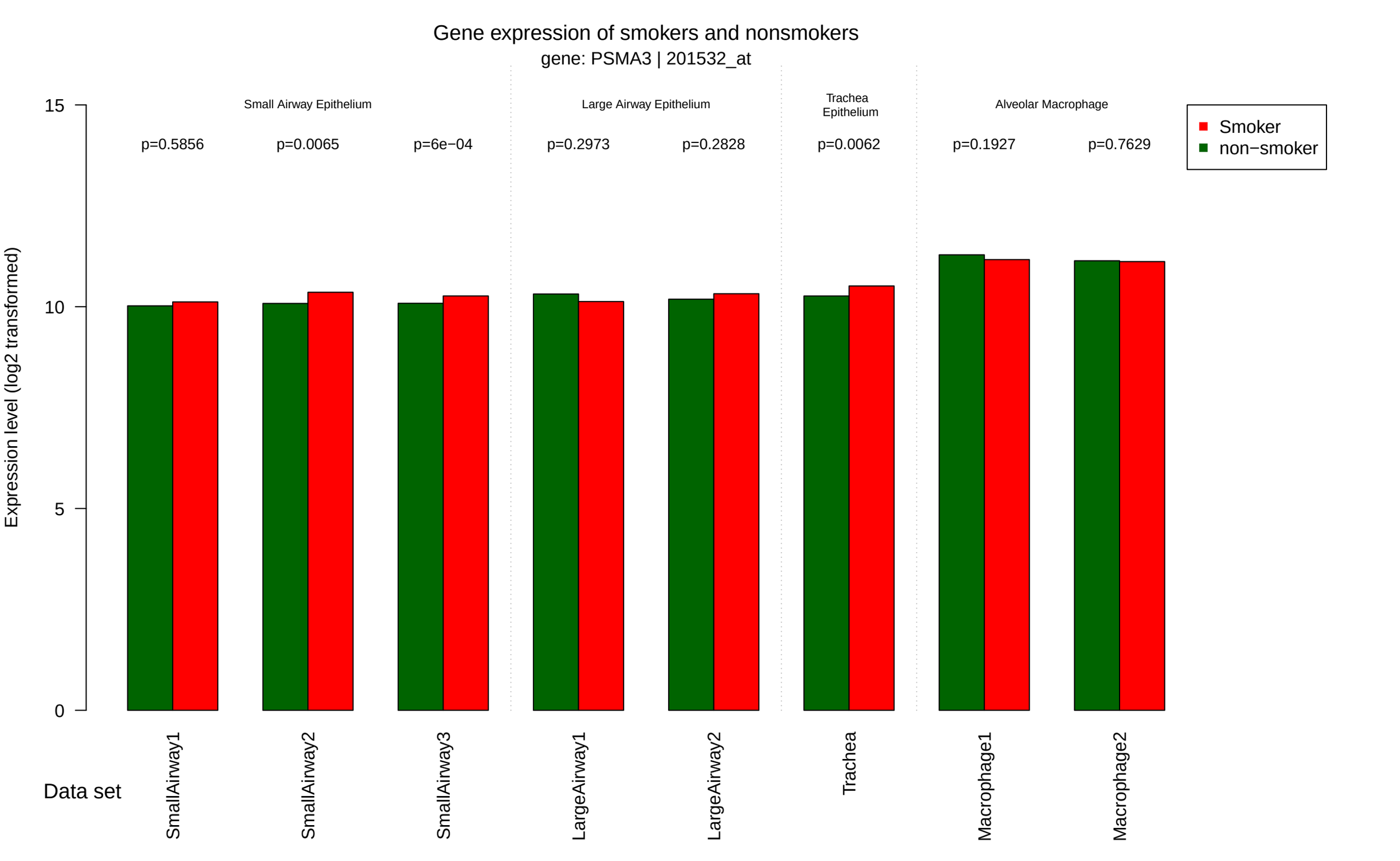
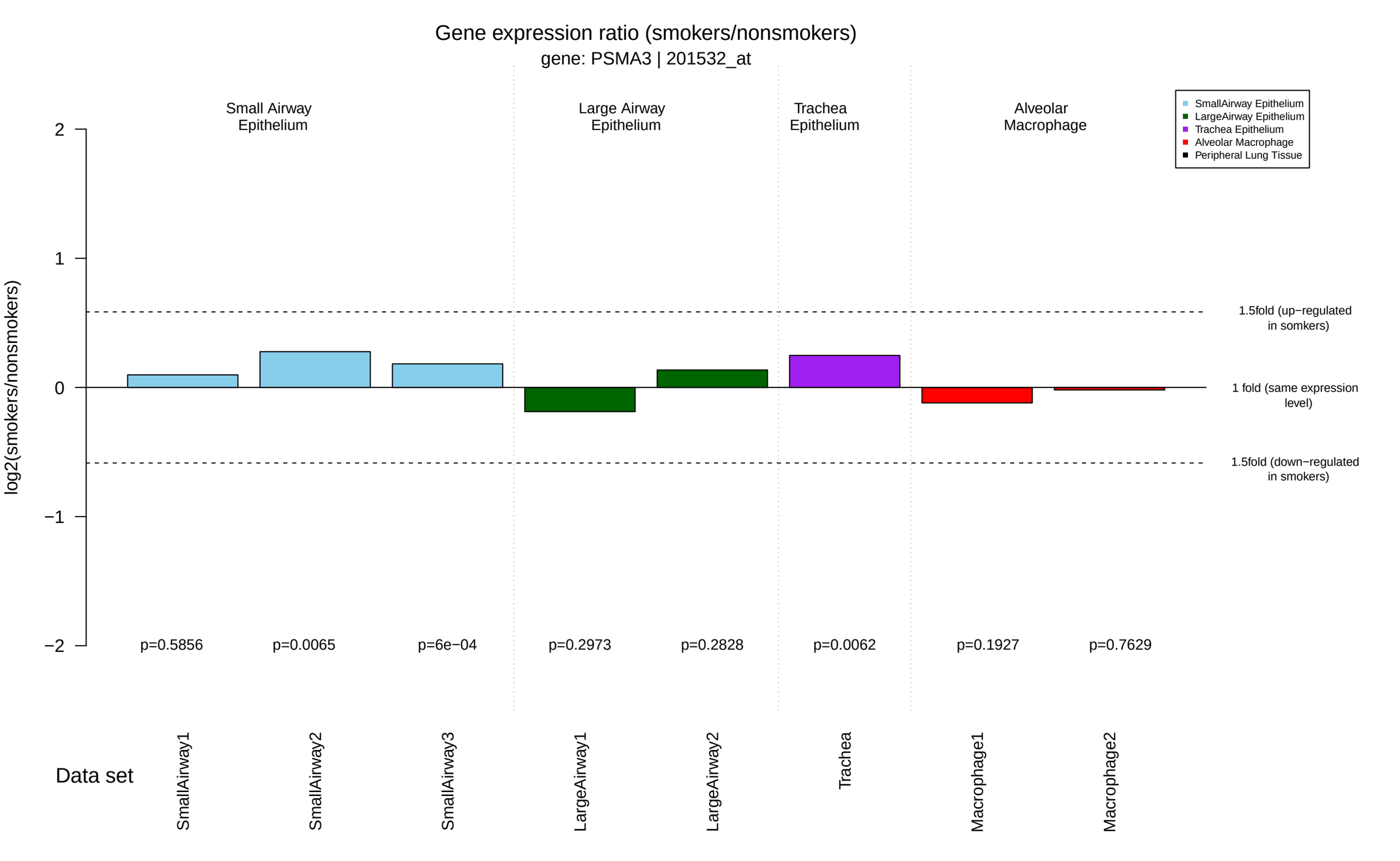
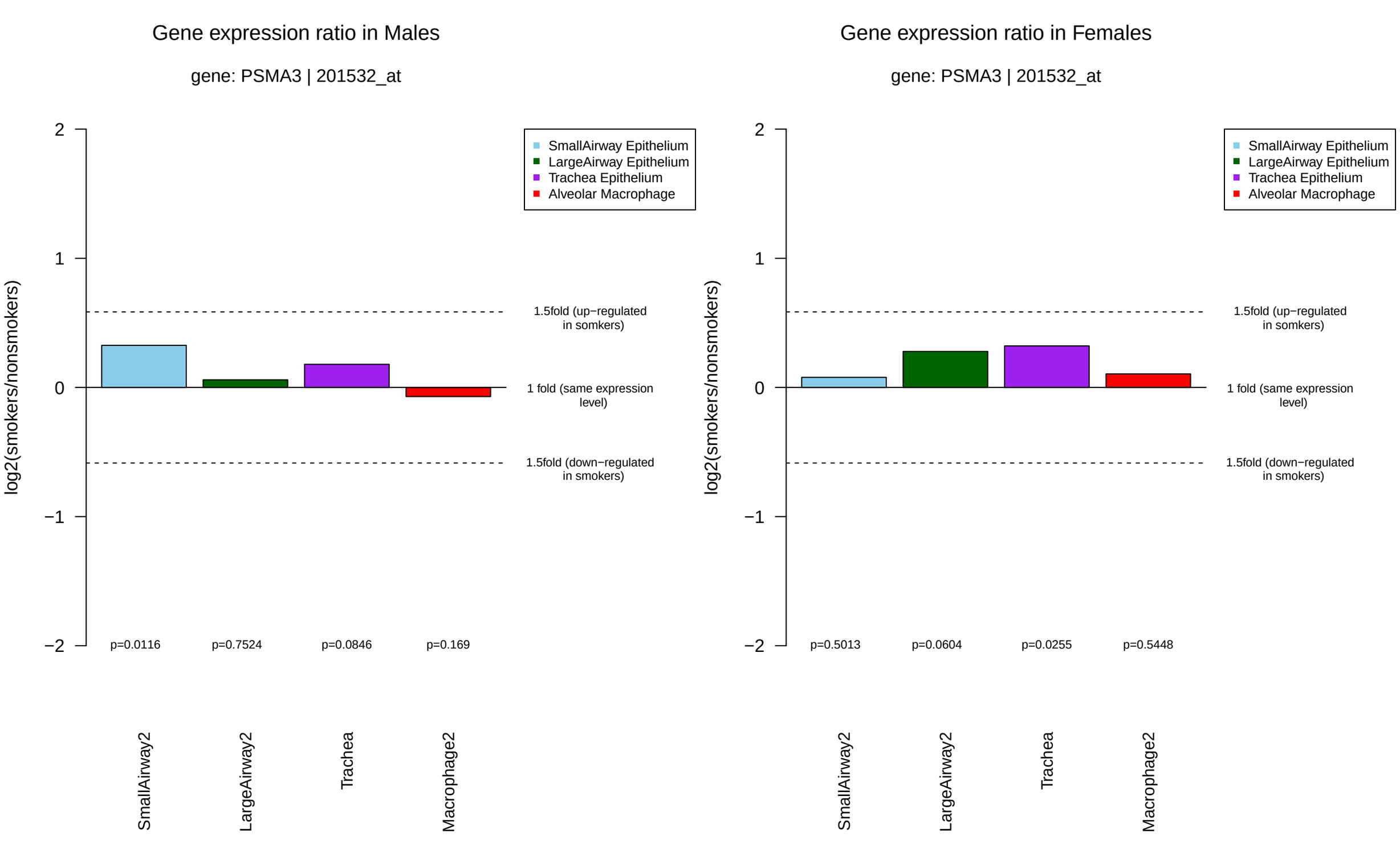
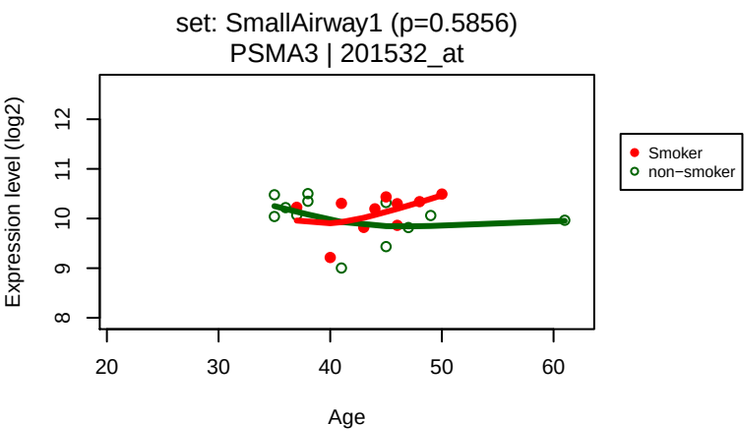
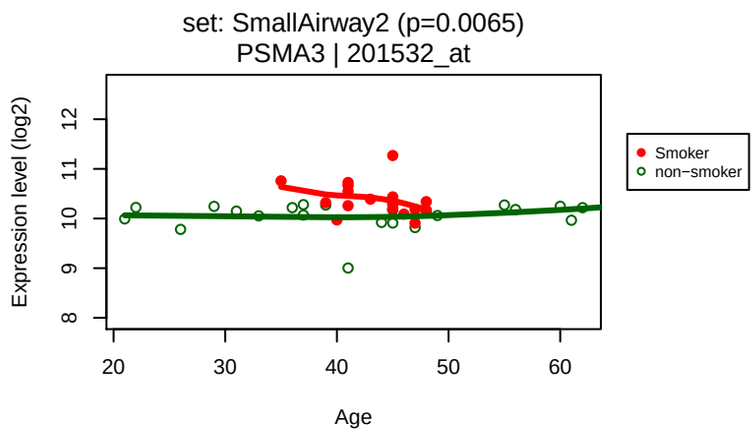
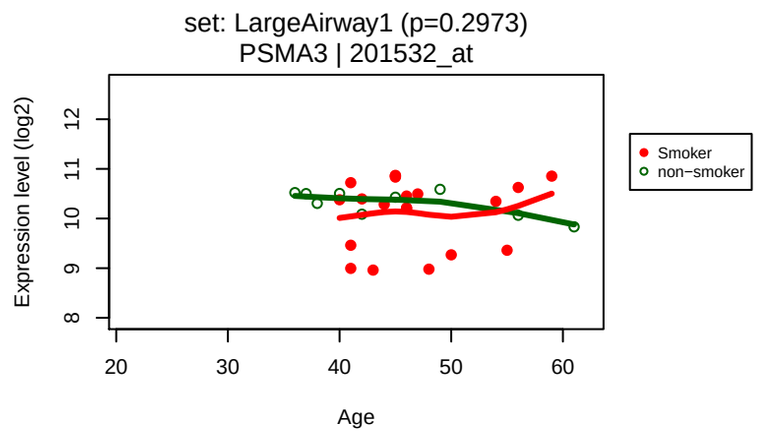

Gene expression of smokers and nonsmokers
gene: PSMA3 | 201532_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5856
p=0.0065
p=6e−04
p=0.2973
p=0.2828
p=0.0062
p=0.1927
p=0.7629
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

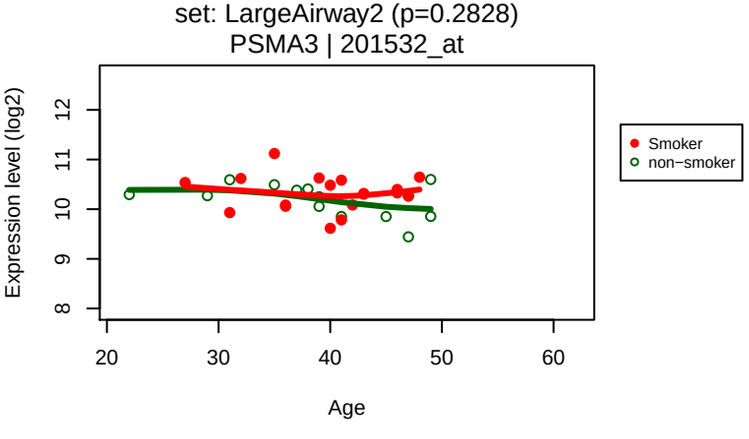

Gene expression ratio (smokers/nonsmokers)
gene: PSMA3 | 201532_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5856
p=0.0065
p=6e−04
p=0.2973
p=0.2828
p=0.0062
p=0.1927
p=0.7629
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

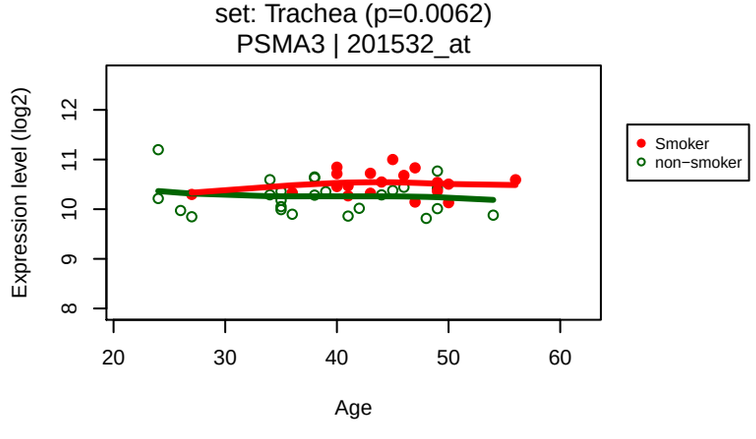

Gene expression ratio in Males
gene: PSMA3 | 201532_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0116
p=0.7524
p=0.0846
p=0.169
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PSMA3 | 201532_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5013
p=0.0604
p=0.0255
p=0.5448
SmallAirway2
LargeAirway2
Trachea
Macrophage2

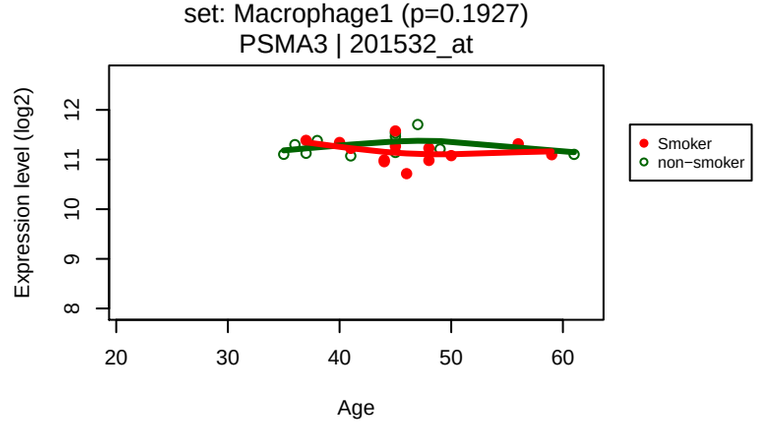

set: SmallAirway1 (p=0.5856)
PSMA3 | 201532_at
Expression level (log2)
Age
Smoker
non−smoker

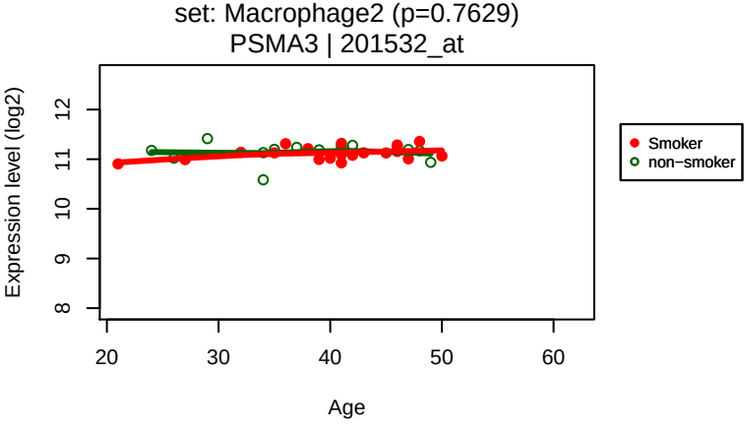

set: SmallAirway2 (p=0.0065)
PSMA3 | 201532_at
Expression level (log2)
Age
Smoker
non−smoker

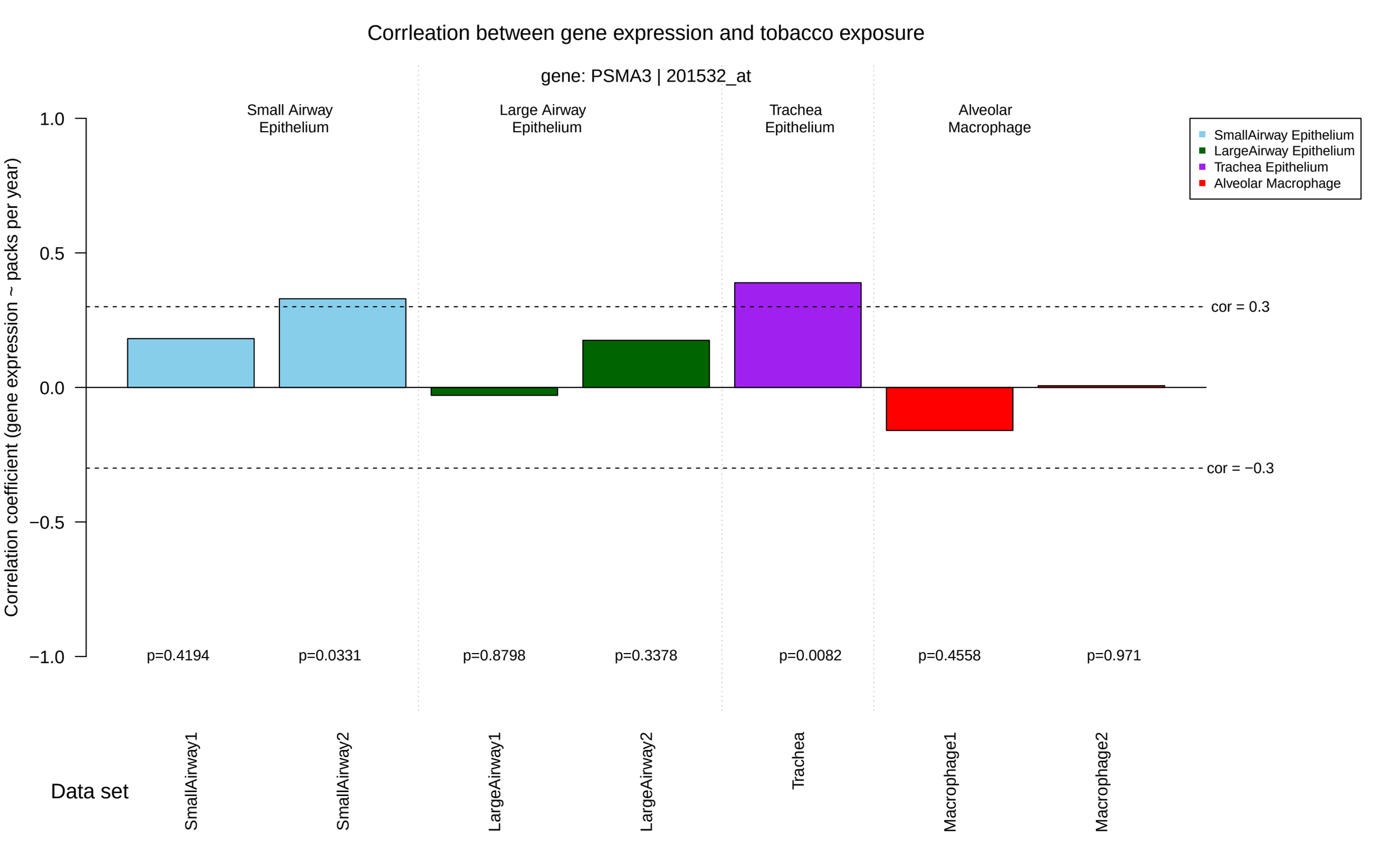

set: LargeAirway1 (p=0.2973)
PSMA3 | 201532_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2828)
PSMA3 | 201532_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0062)
PSMA3 | 201532_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1927)
PSMA3 | 201532_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7629)
PSMA3 | 201532_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PSMA3 | 201532_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4194
p=0.0331
p=0.8798
p=0.3378
p=0.0082
p=0.4558
p=0.971
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2